
TN 15.3.0.1: Collecting and Interpreting Monitoring Information of gen4 Devices

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Contents

1	Context and Challenge: Finding Root Causes of Non-Systematic Issues	1
2	Solution: Client-Based System Monitoring	1
2.1	Permanently Available Status Information	2
2.2	Status Information Relevant after Configuration	2
2.3	Status Information of the Acquisition Engine	3
2.4	Status Information Related to Measurement Data	3
3	Document History	4

1 Context and Challenge: Finding Root Causes of Non-Systematic Issues

HelilInspect H8, H9, or heliCam C4 devices may suddenly, sporadically, or gradually fail or deliver poor quality data. Hardware defects or environmental issues, including interactions on their interfaces, are possible root causes.

The devices and software tools do, for the time being, not provide accessible log files or interactive monitoring of internal or external operating conditions. Finding root causes of such issues, therefore, requires high debugging efforts.

2 Solution: Client-Based System Monitoring

Debugging of issues is simplified or made possible in the first place by collecting and possibly monitoring status information from the device. This should be implemented in the client (consumer) application that controls the heliCam/heliInspect device.

The tables below point at useful status information. Client applications read it through the standard camera interface at the configuration stage, after performing a measurement, or in case of an exception. Depending on the type of information and on user requirements, the application may implement monitoring and/or logging.

2.1 Permanently Available Status Information

The client application can read the device and status information listed below at any time as long as it has an open connection to the device.

Feature	Value [expected / healthy]
DeviceFirmwareVersion	camera embedded SW version (format e.g. 1.x.y)
DeviceFPGAfwVersion	camera fpga firmware version in cryptic format
HeliDriverFirmwareVersion	heliDriver firmware version (format: e.g. 1.x)
HeliDriverStatus	status of the heliDriver in cryptic format
DeviceTemperature [DeviceTemperatureSelector= <i>Processor</i>]	< 80
DeviceMonitorActualValue [DeviceMonitorSelector = <i>Vdd-Voltage</i>]	between 21.5 and 26.5
DeviceMonitorActualValue [DeviceMonitorSelector = <i>Vdd-Current</i>]	< 1.5
DeviceMonitorActualValue [DeviceMonitorSelector= <i>UserVdd-Voltage</i>]	between 21.5 and 26.5
DeviceMonitorActualValue [DeviceMonitorSelector= <i>UserVdd-Current</i>]	< 0.15

Note: Refer to the feature description document for data types and detailed information about the features mentioned.

Note: Some features are sensitive to values of selector features. Refer to the programmer's manual for more information on the concept.

Note: Providing logged information in support requests to heliotis is very helpful. In particular, please contact heliotis for interpreting information in cryptic format.

2.2 Status Information Relevant after Configuration

The status information below is readable any time but its content is sensitive to configuration features. The client application may log it after configuring the device.

Feature	Value [expected / healthy]
ActualVerticalSpacing FrameRate	near TargetVerticalSpacing
	H8,H9,C4 (542x512 pixel) devices: < 4000 H8M, H9M, C4M (1.1M pixel) devices: < 1000
RecordingNFrame TriggerPosition	approximately: $\frac{ScanRange}{ActualVerticalSpacing}$ defined by ScanMode , ScanPosition , ScanRange
LightBrightness [LightControllerSelector = <i>Light-Controller0</i>]	10% to 100%

Note: FrameRate maximum values are approximate. The exact maximum depends on further settings.

2.3 Status Information of the Acquisition Engine

The following status information represents the internal state of the acquisition engine. In case of a timeout during recording, the client application could find out why the acquisition engine got stuck.

Feature	Value
AcquisitionStatus [AcquisitionStatusSelector = <i>FrameTriggerWait</i>]	<i>True</i> : state active, <i>False</i> : state inactive
AcquisitionStatus [AcquisitionStatusSelector = <i>RecordingTriggerWait</i>]	<i>True</i> : state active, <i>False</i> : state inactive
AcquisitionStatus [AcquisitionStatusSelector = <i>FrameActive</i>]	<i>True</i> : state active, <i>False</i> : state inactive
AcquisitionStatus [AcquisitionStatusSelector = <i>RecordingActive</i>]	<i>True</i> : state active, <i>False</i> : state inactive
TransferStatus [TransferStatusSelector = <i>Streaming</i>]	<i>True</i> : state active, <i>False</i> : state inactive
StageStatus [StageStatusSelector = <i>InMotion</i>]	<i>True</i> : state active, <i>False</i> : state inactive

Note: All possible acquisition engine states are documented in the appendix of the feature description.

2.4 Status Information Related to Measurement Data

The following status information is related to measurement data. It should be read after a successful measurement and always relates to the latest measurement.

Feature	Value (expected / healthy)
LightCurrentMeasured [LightControllerSelector = <i>LightController0</i>]	$LightCurrentRating[.] * \frac{LightBrightness[.]}{100\%}$
ChunkZTagCount	equal to RecordingNFrame
ChunkZTagValue [ChunkZTagSelector = <i>1</i>]	$\frac{TriggerPosition}{EncoderResolution[Camera]}$
ChunkProcessingString	(.json string) dump from on-camera data processing
ChunkConfigString	(.json string) dump of camera configuration

Note: Make sure the illumination is enabled when reading the value of **LightCurrentMeasured**. Typical values for LED current rating are 1A (H8) or 2A (H9).

Much of the measurement-related information are transmitted as GenTL ChunkData along with measurement data. The host application must enable transmission by setting the following configuration features already before the measurement, e.g. when initializing the camera:

Listing 2.1: Enable relevant chunk data

```
ChunkModeActive = True
ChunkSelector = "ZTagCount"
ChunkEnable = True
ChunkSelector = "ZTagValue"
ChunkEnable = True
ChunkSelector = "ProcessingString"
ChunkEnable = True
ChunkSelector = "ConfigString"
ChunkEnable = True
```

Note: Some of the debugging chunk data like *ConfigurationString* could significantly increase the measurement cycle time. General enabling is not recommended.

3 Document History

Version	Date	Changes	Responsible
15.3.0.0	14-11-2022	initial version	cl
15.3.0.1	01-03-2024	review/correction	sm